Your 8-hour Chemistry rescue plan

Everything that can show up in the Kerala **Plus Two (Class 12) Chemistry SAY exam 2026** — every chapter's important topics explained in plain language, how each topic gets asked, and practice questions with full model answers written the way the examiner wants them. Built around the real **60-mark theory pattern**.

📘 **12 active chapters** (+4 reportedly excluded) ⏱ **2 hrs + 15 min cool-off** ✍️ **1 / 2 / 3 / 4-mark questions**

✅ **Pass mark: 18 / 60 (30%)** 🚫 **No negative marking**

⏰ How to spend your 8 hours

1. **0:00–0:20** Read the exam-pattern + answer-writing tips at the bottom first. Know how marks are won.
2. **0:20–3:30** **Organic chemistry block** (Ch 10–13) — highest weightage, most predictable. Learn name-reactions + distinguishing tests cold.
3. **3:30–5:00** **Physical block** — Electrochemistry, Chemical Kinetics, Solutions. Practise the numericals; the formulas repeat.
4. **5:00–6:30** **Inorganic block** — Coordination Compounds (CFT!), d & f-block ($KMnO_4$/$K_2Cr_2O_7$).
5. **6:30–7:30** Biomolecules, Polymers, Chemistry in Everyday Life — short, scoring, easy to memorise.
6. **7:30–8:00** Run **Quiz me**, re-read every "How it's asked" box, sleep if you can.

🎯 If you only have time for the big scorers

1. **Aldehydes/Ketones/Acids** — Tollens, Fehling, Iodoform, Aldol, Cannizzaro.
2. **Coordination Compounds** — IUPAC naming, CFT splitting, magnetic moment.
3. **Electrochemistry** — Nernst equation + EMF numericals.
4. **Chemical Kinetics** — first-order numericals, half-life, Arrhenius.
5. **Alcohols/Phenols/Ethers** — phenol acidity, Reimer–Tiemann, Williamson.
6. **Amines** — basicity order, diazonium reactions, distinguishing 1°/2°/3°.
7. **d & f-block** — $KMnO_4$, $K_2Cr_2O_7$, lanthanoid contraction.
8. **Solutions** — colligative properties + van't Hoff factor.

These eight chapters alone typically cover well over half the paper. Master them first.

⚡ Quick Revision — study this if you're short on time

The 12 in-syllabus chapters, condensed. Read the **must-know** points, memorise the **key reactions**, and make sure you can answer the 🔁 **sure questions**. This alone gets you well past the pass mark. Full explanations are in the chapters below.

Ch 12 Aldehydes, Ketones & Carboxylic Acids

MUST KNOW

- Carbonyl C is δ+ → nucleophilic addition. Reactivity aldehyde > ketone (less +I, less steric).
- **Tollens'** → aldehyde gives silver mirror, ketone does NOT.
- **Fehling's** → aliphatic aldehyde gives red-brown Cu_2O; ketones & aromatic aldehydes do NOT.
- **Iodoform (I_2/NaOH)** positive for CH_3CHO, methyl ketones (CH_3CO–), ethanol, $CH_3CH(OH)$– → yellow CHI_3.
- **Aldol**: has α-H + dil. base → β-hydroxy carbonyl → α,β-unsaturated. **Cannizzaro**: NO α-H (HCHO, C_6H_5CHO) + conc. base → alcohol + acid salt.
- Acid strength: EWG raises it → $Cl_3CCOOH > Cl_2CHCOOH > ClCH_2COOH > CH_3COOH$.

KEY REACTIONS / FORMULAS

```
Clemmensen (Zn-Hg/HCl) & Wolff–Kishner (NH₂NH₂/KOH): C=O → CH₂
```

SURE QUESTIONS

🔁 Distinguish ethanal vs propanone (Tollens'), propanal vs propanone (Fehling's)

🔁 Cannizzaro of benzaldehyde — why it, not acetaldehyde

🔁 Why aldehydes more reactive than ketones

Ch 11 Alcohols, Phenols & Ethers

MUST KNOW

- **Phenol > alcohol acidity**: phenoxide is resonance-stabilised. –NO_2 at o/p raises acidity (picric acid strong).
- Oxidation: 1° → aldehyde → acid; 2° → ketone; 3° resists.
- **Lucas test**: 3° immediate turbidity, 2° ~5 min, 1° no reaction (cold).
- Phenol + neutral $FeCl_3$ → violet colour (test for phenol).

KEY REACTIONS / FORMULAS

```
Williamson: alkoxide + 1° alkyl halide → ether (SN2; 1° avoids elimination)
```

```
Kolbe: Na-phenoxide + CO₂ → salicylic acid
```

```
Reimer–Tiemann: phenol + CHCl₃ + KOH → salicylaldehyde (–CHO ortho)
```

SURE QUESTIONS

🔁 Why phenol more acidic than ethanol + nitro effect

🔁 Reimer–Tiemann / Kolbe reaction

🔁 Williamson synthesis — why a primary halide

Ch 13 Amines

MUST KNOW

- **Basicity (aqueous, simple alkyl): 2° > 1° > 3°**. Aniline weakest (lone pair into ring).
- **Carbylamine test** (R-NH_2 + $CHCl_3$ + alc.KOH → foul isocyanide): only 1° amines.
- Hinsberg's reagent distinguishes 1°/2°/3°.

KEY REACTIONS / FORMULAS

```
Gabriel → pure 1° amine (no aromatic amines)
```

```
Hoffmann bromamide (R-CONH₂ + Br₂/NaOH) → 1° amine, one C fewer
```

```
Diazotisation: Ar-NH₂ + NaNO₂ + HCl (0–5°C) → Ar-N₂⁺; then Sandmeyer / coupling (azo dyes)
```

SURE QUESTIONS

🔁 Order of basicity: aniline, ammonia, methylamine + reason

🔁 Synthetic uses of diazonium salts

🔁 Distinguish 1°/2°/3° amine

Ch 10 Haloalkanes & Haloarenes

MUST KNOW

- **SN2**: 1 step, rate=k[RX][Nu], inversion (Walden), favoured by 1°.
- **SN1**: 2 steps (carbocation), rate=k[RX], racemisation, favoured by 3°.
- Haloarenes less reactive: C–X partial double bond (resonance), sp^2 C, ring repels nucleophile.
- Saytzeff: more-substituted (stable) alkene is the major elimination product.

KEY REACTIONS / FORMULAS

```
R-OH → R-Cl best with SOCl₂ (by-products SO₂, HCl are gases)
```

SURE QUESTIONS

🔁 Compare SN1 and SN2

🔁 Why chlorobenzene less reactive than chloromethane

🔁 Saytzeff rule

Ch 9 Coordination Compounds

MUST KNOW

- Chelate = ring from polydentate ligand (extra stable). Ambidentate = binds 2 ways (NO_2^-, SCN^-).
- IUPAC: ligands alphabetical, anionic end -o (chlorido, cyanido), complex anion ends -ate, ox. state in Roman.
- **CFT octahedral**: t_2g (low) + e_g (high), gap Δ_0. Tetrahedral reversed, always high-spin.
- Spectrochemical (weak → strong): $I^-<F^-<H_2O<NH_3<en<CN^-\approx CO$. Strong field → low-spin.
- Colour = d–d transition; magnetism from unpaired e^-, $\mu=\sqrt{n(n+2)}$ BM.

KEY REACTIONS / FORMULAS

```
[Co(NH₃)₆]³⁺ low-spin diamagnetic ; [CoF₆]³⁻ high-spin paramagnetic
```

SURE QUESTIONS

🔁 CFT: why $[Co(NH_3)_6]^{3+}$ diamagnetic, $[CoF_6]^{3-}$ paramagnetic

🔁 IUPAC names of given complexes

🔁 Define chelate / ambidentate ligand

Ch 8 d- and f-Block Elements

MUST KNOW

- Transition metals: variable oxidation states, coloured ions, paramagnetism, catalysts, complexes/alloys.
- $\mu = \sqrt{n(n+2)}$ BM (n = unpaired e^-). e.g. Fe^{2+} (d^6, n=4) → 4.9 BM.
- **$KMnO_4$**: Mn^{7+} → Mn^{2+} (5 e^-) in acid; strong oxidiser. **$K_2Cr_2O_7$**: Cr^{6+} → Cr^{3+} (6 e^-); orange↔yellow with base.
- **Lanthanoid contraction**: 4f poor shielding → radii ↓ → Zr ≈ Hf (hard to separate).

KEY REACTIONS / FORMULAS

$$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$$

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$$

SURE QUESTIONS

🔁 Lanthanoid contraction + consequences

🔁 Spin-only magnetic moment of a given ion

🔁 Why transition metals show variable oxidation states / colour

Ch 3 Electrochemistry

MUST KNOW

- E°cell = E°cathode − E°anode. +ve E° → spontaneous ($\Delta G = -nFE$).
- **Nernst (298K)**: $E = E° - (0.0591/n)\cdot\log Q$.
- On dilution: conductivity κ ↓ but molar conductivity Λm ↑.
- Kohlrausch's law → Λ°m of weak electrolyte and degree of dissociation. 1 F = 96500 C.

KEY REACTIONS / FORMULAS

$$\log K_c = nE°/0.0591$$

$$\Lambda_m = \kappa \times 1000/c$$

$$w = MIt/(nF)$$

SURE QUESTIONS

🔁 EMF of a cell by Nernst equation (numerical)

🔁 Kohlrausch's law + application

🔁 Why Λm increases but κ decreases on dilution

Ch 4 Chemical Kinetics

MUST KNOW

- Order = sum of powers in experimental rate law (0/fraction possible). Molecularity = whole number, elementary step.
- **First order**: $k=(2.303/t)\log([R]_0/[R])$; $t_{1/2}=0.693/k$ (independent of conc.).
- Zero order: $k=([R]_0-[R])/t$. Pseudo first order: ester hydrolysis (water excess).
- Arrhenius: $k=Ae^{(-Ea/RT)}$; rate ↑ with T; catalyst lowers Ea.

KEY REACTIONS / FORMULAS

```
log(k₂/k₁) = (Ea/2.303R)(1/T₁ − 1/T₂)
```

SURE QUESTIONS

🔁 First-order numerical (find t, k or amount left)

🔁 Order vs molecularity

🔁 Pseudo first-order reaction + example

Ch 2 Solutions

MUST KNOW

- Molarity (temp-dependent) vs molality (temp-independent, used for colligative).
- Henry: $p = K_H \cdot x$. Raoult: $p_A = p°_A \cdot x_A$.
- +ve deviation (ethanol+acetone, min-boiling azeotrope); −ve deviation (chloroform+acetone, max-boiling).
- 4 colligative properties; van't Hoff i >1 (dissociation), <1 (association).

KEY REACTIONS / FORMULAS

```
ΔTb=iKb·m ; ΔTf=iKf·m ; π=iCRT
```

```
M = (iKf·w_B·1000)/(ΔTf·w_A)
```

SURE QUESTIONS

🔁 Molar mass from ΔTf / ΔTb / osmotic pressure (numerical)

🔁 Henry's / Raoult's law + application

🔁 +ve vs −ve deviation with example

Ch 14 Biomolecules

MUST KNOW

- Carbohydrates: mono (glucose, fructose), di (sucrose non-reducing; maltose/lactose reducing), poly (starch, cellulose, glycogen).
- Proteins: α-amino acids + peptide (–CO–NH–) bonds; zwitterion; denaturation loses 2°/3° structure (boiled egg).

- DNA: deoxyribose, A-T-G-C, double helix. RNA: ribose, A-U-G-C, single strand.
- Deficiency: scurvy (C), rickets (D), night blindness (A).

SURE QUESTIONS

🔁 Differences between DNA and RNA

🔁 Denaturation / peptide bond / zwitterion

🔁 Why sucrose is non-reducing

Ch 15 Polymers

MUST KNOW

- **Addition** (chain, no small molecule lost): polythene, PVC, Teflon, polystyrene.
- **Condensation** (step, lose H_2O): nylon-6,6, terylene, bakelite.
- Monomers: nylon-6,6 = hexamethylenediamine + adipic acid; terylene = ethylene glycol + terephthalic acid; bakelite = phenol + HCHO; Buna-S = butadiene + styrene.
- Vulcanisation = rubber + sulphur (cross-links → stronger, more elastic).

SURE QUESTIONS

🔁 Addition vs condensation polymer + examples

🔁 Monomers of nylon-6,6 / terylene / bakelite

🔁 Vulcanisation / a biodegradable polymer

Ch 16 Chemistry in Everyday Life

MUST KNOW

- Drug classes: antacids (ranitidine), antihistamines, tranquilizers, analgesics (aspirin), antibiotics (penicillin).
- **Antiseptic** = living tissue (Dettol) vs **disinfectant** = non-living surface (conc. phenol). Phenol is both, by concentration.
- Soaps fail in hard water (Ca^{2+}/Mg^{2+} scum); detergents work (soluble Ca/Mg salts).
- Cleansing = micelle (tail into grease, ionic head into water).

SURE QUESTIONS

🔁 Antiseptics vs disinfectants + examples

🔁 Soaps vs detergents / why soaps fail in hard water

🔁 Cleansing action of soap (micelle)

⚠️ Reportedly deleted from the 2026 syllabus — hidden

One detailed source lists these **4 chapters as removed** from the Kerala Plus Two Chemistry 2025-26 syllabus (they match the NCERT 2023 rationalisation): **The Solid State · Surface Chemistry · Isolation of Elements (Metallurgy) · The p-Block Elements**. They are **hidden** so you don't waste time, and left out of the quiz and PDF.

Other sources say "no change," so confirm on dhsekerala.gov.in before skipping them. Nothing is deleted permanently — one tap brings them back.

Hidden from study, quiz and PDF.

Ch 2 Solutions

HIGH

💡 A reliable source of one full numerical (colligative property) plus a definition. Learn the four colligative formulas and van't Hoff factor.

IMPORTANT TOPICS (EXPLAINED)

Concentration terms

- **Molarity (M)** = mol solute / L solution (temperature-dependent).
- **Molality (m)** = mol solute / kg solvent (temperature-independent — preferred for colligative work).
- **Mole fraction (x)** = mol component / total mol.
- **ppm, mass %, volume %.**

Henry's law & Raoult's law

Henry's law: the solubility of a gas in a liquid is proportional to its partial pressure: $p = K_H \cdot x$. Higher $K_H \implies$ lower solubility.

Raoult's law: for a solution of two volatile liquids, the partial vapour pressure of each component is proportional to its mole fraction: $p_A = p°_A \cdot x_A$. Total pressure $= p°_A x_A + p°_B x_B$.

Ideal & non-ideal solutions

Ideal: obey Raoult's law at all concentrations, $\Delta H_{mix} = 0$, $\Delta V_{mix} = 0$ (e.g. benzene + toluene).

Positive deviation: weaker A–B forces, higher vapour pressure (ethanol + acetone) → minimum boiling azeotrope.

Negative deviation: stronger A–B forces, lower vapour pressure (chloroform + acetone) → maximum boiling azeotrope.

Colligative properties

Properties that depend on the **number** of solute particles, not their nature:

1. Relative lowering of vapour pressure
2. Elevation of boiling point (ΔT_b)
3. Depression of freezing point (ΔT_f)
4. Osmotic pressure (π)

Used to find molar mass of solute.

van't Hoff factor (i) & abnormal molar mass

i = (observed colligative effect)/(normal effect) = (no. of particles after dissociation/association)/(particles before).

- Electrolytes that dissociate: $i > 1$ (NaCl ≈ 2, $BaCl_2$ ≈ 3).
- Solutes that associate (e.g. acetic acid in benzene → dimer): $i < 1$.

MUST-KNOW FORMULAS & REACTIONS

```
Relative lowering of VP: (p° − p)/p° = x_solute
```

```
Elevation of boiling point: ΔT_b = i · K_b · m
```

```
Depression of freezing point: ΔT_f = i · K_f · m
```

```
Osmotic pressure: π = i · C · R · T (C in mol/L, R = 0.0821 L atm K⁻¹mol⁻¹)
```

```
Molar mass from depression: M = (i · K_f · w_B · 1000)/(ΔT_f · w_A)
```

HOW QUESTIONS COME FROM THIS CHAPTER

▶ 2-mark: state Henry's law / Raoult's law with one application.

▶ 3-mark: numerical — calculate molar mass from ΔT_b, ΔT_f or osmotic pressure.

▶ 2-mark: positive vs negative deviation with one example each.

▶ 1-mark: which colligative property is used to measure very large molar masses? (osmotic pressure).

PRACTICE QUESTIONS WITH MODEL ANSWERS

3 marks **1.0 g of a non-electrolyte solute dissolved in 50 g of benzene lowered the freezing point by 0.40 K. K_f for benzene = 5.12 K kg mol^{-1}. Find the molar mass of the solute.**

MODEL ANSWER

Formula: $M = (K_f \cdot w_B \cdot 1000)/(\Delta T_f \cdot w_A)$ ($i = 1$, non-electrolyte).

$M = (5.12 \times 1.0 \times 1000)/(0.40 \times 50) = 5120/20 =$ **256 g/mol**.

2 marks **Define colligative property. Name the four colligative properties.**

MODEL ANSWER

A **colligative property** depends only on the *number* of solute particles in a solution and not on their chemical nature.

The four are: (1) relative lowering of vapour pressure, (2) elevation of boiling point, (3) depression of freezing point, (4) osmotic pressure.

2 marks **Why does a solution of chloroform and acetone show negative deviation from Raoult's law?**

MODEL ANSWER

Acetone and chloroform form **intermolecular hydrogen bonds** ($Cl_3C–H\cdots O=C$), so the A–B attractions are stronger than the A–A and B–B attractions in the pure liquids. This lowers the escaping tendency, so the vapour pressure is lower than predicted by Raoult's law → negative deviation (and a maximum-boiling azeotrope).

1 mark **What is the value of van't Hoff factor for NaCl in water (assuming complete dissociation)?**

MODEL ANSWER

$i = 2$ ($NaCl \rightarrow Na^+ + Cl^-$).

Ch 3 Electrochemistry

HIGH

💡 Top-priority physical chapter. Expect a Nernst-equation or EMF numerical plus conductance theory. The formulas repeat every year.

IMPORTANT TOPICS (EXPLAINED)

Galvanic (voltaic) cell & electrode potential

A galvanic cell converts chemical energy into electrical energy. **Oxidation at anode (−), reduction at cathode (+).** Standard hydrogen electrode (SHE) is the reference, $E° = 0$ V.

Cell notation: anode **|** anode solution || cathode solution **|** cathode.

Cell EMF & Nernst equation

$E°_{cell} = E°_{cathode} - E°_{anode}$ (both as reduction potentials). The Nernst equation gives EMF at non-standard conditions.

EMF, Gibbs energy & equilibrium constant

$\Delta G° = -nFE°_{cell}$ links electrochemistry to thermodynamics. A positive $E°_{cell} \implies$ negative $\Delta G° \implies$ spontaneous. Also $E°_{cell}$ relates to the equilibrium constant K_c.

Conductance & Kohlrausch's law

Conductivity (κ) decreases with dilution; **molar conductivity (Λ_m)** increases with dilution. For strong electrolytes Λ_m vs $\sqrt{c}$ is linear (Debye–Hückel–Onsager). **Kohlrausch's law:** limiting molar conductivity is the sum of ionic contributions — used to find $\Lambda°_m$ of weak electrolytes.

Electrolysis & Faraday's laws

First law: mass deposited ∝ charge ($w = Z \cdot I \cdot t$). **Second law:** for the same charge, masses are proportional to equivalent masses. 1 Faraday = 96500 C = charge of 1 mole of electrons.

Batteries, fuel cells & corrosion

Primary (dry cell, mercury) — not rechargeable. **Secondary** (lead storage, Ni–Cd) — rechargeable. **Fuel cell** (H_2–O_2) — continuous, pollution-free. **Corrosion** of iron is an electrochemical process; prevented by galvanising, painting, sacrificial protection.

MUST-KNOW FORMULAS & REACTIONS

$$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$$

$$\text{Nernst (298 K): } E_{cell} = E^\circ_{cell} - (0.0591/n) \cdot \log Q$$

$$\Delta G^\circ = -nFE^\circ_{cell} \text{ (F = 96500 C)}$$

$$\log K_c = (n \cdot E^\circ_{cell})/0.0591$$

$$\text{Molar conductivity: } \Lambda_m = (\kappa \times 1000)/c \text{ (c in mol/L)}$$

$$\text{Kohlrausch: } \Lambda^\circ_m = \nu_+ \cdot \lambda^\circ_+ + \nu_- \cdot \lambda^\circ_-$$

$$\text{Faraday's first law: } w = (M \cdot I \cdot t)/(n \cdot F)$$

HOW QUESTIONS COME FROM THIS CHAPTER

- ▶ 3-mark: calculate E_{cell} using the Nernst equation given concentrations.
- ▶ 3-mark: calculate Λ°_m of a weak electrolyte (e.g. acetic acid) using Kohlrausch's law.
- ▶ 2-mark: differences between primary and secondary cells / write electrode reactions of a given cell.
- ▶ 1-mark: value of 1 Faraday / which electrode is the anode in a galvanic cell.

PRACTICE QUESTIONS WITH MODEL ANSWERS

3 marks **Calculate the EMF of the cell $Zn|Zn^{2+}(0.1\ M)||Cu^{2+}(1.0\ M)|Cu$ at 298 K. Given $E^\circ(Zn^{2+}/Zn) = -0.76$ V, $E^\circ(Cu^{2+}/Cu) = +0.34$ V.**

MODEL ANSWER

E°_{cell} = $E^\circ_{cathode} - E^\circ_{anode} = 0.34 - (-0.76) =$ **1.10 V**.

$n = 2$. Reaction: $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$, $Q = [Zn^{2+}]/[Cu^{2+}] = 0.1/1.0 = 0.1$.

$E_{cell} = 1.10 - (0.0591/2)\cdot\log(0.1) = 1.10 - (0.02955)(-1) = 1.10 + 0.0296 =$ **1.13 V**.

2 marks **State Kohlrausch's law and give one application.**

MODEL ANSWER

Kohlrausch's law: at infinite dilution, the limiting molar conductivity of an electrolyte is the sum of the individual limiting molar conductivities of its cations and anions.

Application: calculating $\Lambda°_m$ of a weak electrolyte (like acetic acid) which cannot be found by extrapolation; also for finding the degree of dissociation ($\alpha = \Lambda_m/\Lambda°_m$).

2 marks **Why does molar conductivity increase on dilution while conductivity decreases?**

MODEL ANSWER

On dilution the number of ions per unit volume falls, so **conductivity (κ)** — which measures conducting power per unit volume — decreases. But **molar conductivity** accounts for all ions from one mole of electrolyte; on dilution more of the electrolyte dissociates and ions move more freely, so Λ_m increases and reaches a maximum ($\Lambda°_m$) at infinite dilution.

1 mark **What is the value of one Faraday of charge?**

MODEL ANSWER

96500 C (charge carried by one mole of electrons).

Ch 4 Chemical Kinetics

HIGH

💡 Numerical-heavy and predictable. The first-order integrated equation and half-life appear almost every year. Learn order vs molecularity cold.

IMPORTANT TOPICS (EXPLAINED)

Rate of reaction & rate law

Rate = change in concentration per unit time. **Rate law**: Rate = $k[A]^x[B]^y$, where x, y are found experimentally. **Order** = x + y (sum of powers). **Molecularity** = number of species in an elementary step (always a whole number ≥ 1).

Order vs molecularity

- Order is experimental, can be 0, fractional, even negative; defined for overall reaction.
- Molecularity is theoretical, a positive integer, defined only for elementary steps.
- The slowest step is the **rate-determining step**.

Integrated rate equations

Zero order: $[R] = [R]_0 - kt$; units of k = mol L^{-1} s^{-1}.

First order: $k = (2.303/t)\cdot\log([R]_0/[R])$; units of k = s^{-1}.

Half-life

First order: $t_{1/2} = 0.693/k$ — independent of initial concentration. **Zero order:** $t_{1/2} = [R]_0/2k$.

Pseudo first order reactions

Reactions that are actually higher order but behave as first order because one reactant (often water) is in large excess and its concentration stays effectively constant. Example: acid hydrolysis of an ester, inversion of cane sugar.

Temperature dependence — Arrhenius equation & collision theory

$k = A\cdot e^{(-Ea/RT)}$. Higher temperature ⟹ more molecules cross the activation-energy barrier ⟹ faster rate (rate roughly doubles per 10 °C). **Activation energy (Ea)** is the minimum energy needed; a catalyst lowers Ea by providing an alternate path.

MUST-KNOW FORMULAS & REACTIONS

```
First order: k = (2.303/t)·log([R]₀/[R])
```

```
First-order half-life: t₁/₂ = 0.693/k
```

```
Zero order: k = ([R]₀ − [R])/t ; t₁/₂ = [R]₀/2k
```

```
Arrhenius: k = A·e^(−Ea/RT)
```

```
log(k₂/k₁) = (Ea/2.303R)·(1/T₁ − 1/T₂)
```

HOW QUESTIONS COME FROM THIS CHAPTER

- ▶ 3-mark: first-order numerical — find k, or time for a given fraction to react, or remaining concentration.
- ▶ 2-mark: difference between order and molecularity.
- ▶ 2-mark: explain pseudo first-order reaction with an example.
- ▶ 1-mark: units of rate constant for a first-order / zero-order reaction.

PRACTICE QUESTIONS WITH MODEL ANSWERS

3 marks **A first-order reaction has a rate constant of 1.15×10^{-3} s^{-1}. How long will it take for 5 g of the reactant to reduce to 3 g?**

MODEL ANSWER

Formula: $k = (2.303/t)\cdot\log([R]_0/[R]) \Longrightarrow t = (2.303/k)\cdot\log([R]_0/[R])$.

$t = (2.303 / 1.15\times10^{-3}) \cdot \log(5/3) = 2002.6 \times \log(1.667) = 2002.6 \times 0.2219 \approx$ **444 s**.

2 marks **Distinguish between order and molecularity of a reaction.**

MODEL ANSWER

- **Order** is the sum of the powers of concentration terms in the experimentally-determined rate law; it can be zero, fractional or integral and applies to the overall reaction.
- **Molecularity** is the number of reacting species in an elementary reaction; it is always a whole number and has meaning only for elementary steps.

2 marks **What is a pseudo first-order reaction? Give one example.**

MODEL ANSWER

A reaction that is not truly first order but follows first-order kinetics because one reactant is present in such large excess that its concentration remains practically constant. **Example:** acid-catalysed hydrolysis of ethyl acetate (water is in excess) — $CH_3COOC_2H_5 + H_2O \rightarrow CH_3COOH + C_2H_5OH$.

1 mark **The half-life of a first-order reaction is independent of what?**

MODEL ANSWER

The **initial concentration** of the reactant.

Ch 8 The d- and f-Block Elements

HIGH

💡 Reliable scorer. Properties of transition metals + the prep/properties of $KMnO_4$ and $K_2Cr_2O_7$ + lanthanoid contraction are near-certain.

IMPORTANT TOPICS (EXPLAINED)

General characteristics of transition elements

- Partly filled d-orbitals in the element or its ions.
- **Variable oxidation states** (because (n−1)d and ns energies are close).
- **Coloured ions** (d–d transitions).
- **Paramagnetism** (unpaired d electrons).
- Good **catalysts**, form **complexes**, **interstitial compounds**, and **alloys**.

Magnetic moment

Spin-only magnetic moment $\mu = \sqrt{n(n+2)}$ BM, where n = number of unpaired electrons. Used to find unpaired electrons / oxidation state.

Potassium permanganate ($KMnO_4$)

Prepared from pyrolusite (MnO_2): fuse with KOH + air/oxidant → K_2MnO_4 (green), then oxidise/electrolyse → $KMnO_4$ (purple). Strong oxidising agent in acidic medium ($Mn^{7+} \rightarrow Mn^{2+}$, gains 5 e^-).

Potassium dichromate ($K_2Cr_2O_7$)

From chromite ore; oxidising agent in acidic medium ($Cr^{6+} \rightarrow Cr^{3+}$, gains 3 e^- per Cr, 6 per dichromate). Orange in acid, converts to yellow chromate in base (interconversion equilibrium).

Lanthanoids & lanthanoid contraction

Lanthanoid contraction: the steady decrease in atomic/ionic radii across the 4f series because the poor shielding by 4f electrons lets the increasing nuclear charge pull the shells inward. **Consequences:** similarity of 4d and 5d elements (Zr–Hf), difficulty in separating lanthanoids, gradual change in basicity.

Actinoids

5f series, all radioactive, show a wider range of oxidation states than lanthanoids; actinoid contraction is greater due to even poorer 5f shielding.

MUST-KNOW FORMULAS & REACTIONS

Spin-only magnetic moment: $\mu = \sqrt{n(n+2)}$ BM (n = unpaired electrons)

$KMnO_4$ in acid: $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$

$K_2Cr_2O_7$ in acid: $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$

Chromate–dichromate: $2CrO_4^{2-} + 2H^+ \rightleftharpoons Cr_2O_7^{2-} + H_2O$

HOW QUESTIONS COME FROM THIS CHAPTER

- ▶ 3-mark: preparation and oxidising action of $KMnO_4$ or $K_2Cr_2O_7$ (with half-reaction).
- ▶ 2-mark: what is lanthanoid contraction and its consequences?
- ▶ 2-mark: calculate spin-only magnetic moment for a given ion (e.g. Fe^{2+}, n=4 $\Longrightarrow$ 4.9 BM).
- ▶ 1-mark: why are transition-metal ions coloured? (d–d transition).

PRACTICE QUESTIONS WITH MODEL ANSWERS

3 marks **What is lanthanoid contraction? Mention two of its consequences.**

MODEL ANSWER

Lanthanoid contraction is the gradual decrease in the atomic and ionic radii of the lanthanoids with increasing atomic number, caused by the poor shielding effect of the 4f electrons, which allows the increasing nuclear charge to draw the electron shells closer.

Consequences: (1) The second and third transition series elements of the same group have almost identical sizes (e.g. Zr and Hf), making them hard to separate. (2) There is a small but steady decrease in the basic strength of the lanthanoid hydroxides across the series.

2 marks **Calculate the spin-only magnetic moment of the Fe^{2+} ion.**

MODEL ANSWER

$Fe^{2+} = [Ar]3d^6 \Longrightarrow$ unpaired electrons n = 4.

$\mu = \sqrt{(n(n+2))} = \sqrt{(4\times6)} = \sqrt{24} \approx$ **4.90 BM**.

2 marks **Why do transition metals show variable oxidation states?**

MODEL ANSWER

Because the energies of the (n−1)d and ns orbitals are very close, electrons from both can take part in bonding. So a transition metal can lose different numbers of electrons and exhibit several oxidation states differing by one (e.g. Fe^{2+}/Fe^{3+}, Mn^{2+} to Mn^{7+}).

1 mark **Why are most transition-metal compounds coloured?**

MODEL ANSWER

Because of **d–d electronic transitions**: an electron absorbs visible light to jump between split d-orbitals, and the complementary colour is transmitted.

Ch 9 Coordination Compounds

HIGH

💡 One of the highest-yield chapters. IUPAC naming, CFT splitting/CFSE, magnetic behaviour and isomerism are exam favourites.

IMPORTANT TOPICS (EXPLAINED)

Werner's theory & basic terms

Ligand — donor species (Lewis base). **Coordination number** — number of donor atoms bonded to the central metal. **Denticity**: monodentate (Cl^-, NH_3), bidentate (en, ox), polydentate (EDTA, hexadentate). **Chelate** — ring formed by a polydentate ligand (extra stability). **Ambidentate** ligand can bind through either of two atoms ($–NO_2$ vs –ONO; –SCN vs –NCS).

IUPAC nomenclature

Rules: name ligands alphabetically before the metal; anionic ligands end in -o (chlorido, cyanido); use di/tri or bis/tris for multiples; cation named first then anion; oxidation state of metal in Roman numerals; complex anion ends in -ate.

Isomerism

Structural: ionisation, linkage, coordination, hydrate. **Stereoisomerism:** geometrical (cis–trans), optical (mirror images).

Valence Bond Theory (VBT)

Explains shape & magnetism via hybridisation: octahedral d^2sp^3 (inner, low-spin) or sp^3d^2 (outer, high-spin); tetrahedral sp^3; square planar dsp^2.

Crystal Field Theory (CFT)

Ligands split the d-orbitals. **Octahedral:** t_2g (lower) and e_g (upper), splitting Δ_0. **Tetrahedral:** reversed, $\Delta_t = (4/9)\Delta_0$ (always high-spin). **Spectrochemical series** ranks ligand field strength: $I^- < Br^- < Cl^- < F^- < OH^- < H_2O < NH_3 <$ en $< CN^- \approx CO$. Strong-field ligands $\Longrightarrow$ large $\Delta \Longrightarrow$ pairing $\Longrightarrow$ low-spin; weak-field $\Longrightarrow$ high-spin. Colour and magnetism follow from the splitting.

Stability & applications

Chelate effect increases stability. Applications: EDTA in water-hardness titration, haemoglobin (Fe), chlorophyll (Mg), vitamin B_{12} (Co), extraction of metals (cyanide process), electroplating.

MUST-KNOW FORMULAS & REACTIONS

```
CFSE (octahedral) = (−0.4 × n_t2g + 0.6 × n_eg) Δ₀ (+ pairing energy terms)
```

```
Δ_t = (4/9) Δ₀ — tetrahedral splitting always smaller ⟹ high spin
```

```
Magnetic moment μ = √(n(n+2)) BM (n = unpaired electrons in the complex)
```

```
Spectrochemical (weak→strong): I⁻
```

HOW QUESTIONS COME FROM THIS CHAPTER

- ▶ 3-mark: write IUPAC name OR formula of given complexes; predict shape/magnetism using CFT.
- ▶ 3-mark: explain why $[Fe(CN)_6]^{4-}$ is low-spin/diamagnetic but $[FeF_6]^{3-}$ is high-spin (spectrochemical series).
- ▶ 2-mark: define ligand, coordination number, chelate, ambidentate ligand.
- ▶ 2-mark: cis–trans / optical isomerism with an example.
- ▶ 1-mark: name the donor atom in NH_3 as a ligand / which is a stronger field ligand, CN^- or H_2O?

PRACTICE QUESTIONS WITH MODEL ANSWERS

3 marks **Using crystal field theory, explain why $[Co(NH_3)_6]^{3+}$ is diamagnetic while $[CoF_6]^{3-}$ is paramagnetic.**

MODEL ANSWER

Both contain Co^{3+} = d^6. The geometry is octahedral, so the d-orbitals split into t_2g and e_g with gap Δ_0.

$[Co(NH_3)_6]^{3+}$: NH_3 is a **strong-field** ligand (high in the spectrochemical series), so Δ_0 is large — larger than the pairing energy. All six electrons pair up in the t_2g set (t_2g^6 e_g^0). No unpaired electrons ⟹ **diamagnetic**, low-spin.

$[CoF_6]^{3-}$: F^- is a **weak-field** ligand, so Δ_0 is small. Electrons fill all five d-orbitals before pairing (t_2g^4 e_g^2), leaving 4 unpaired electrons ⟹ **paramagnetic**, high-spin.

3 marks **Write the IUPAC names of: (a) $K_4[Fe(CN)_6]$ (b) $[Cr(NH_3)_6]Cl_3$ (c) $[Co(en)_3]Cl_3$.**

MODEL ANSWER

(a) **Potassium hexacyanidoferrate(II)**

(b) **Hexaamminechromium(III) chloride**

(c) **Tris(ethane-1,2-diamine)cobalt(III) chloride**

Tip: ligands alphabetical, anionic ligands end -o, complex anion ends -ate, oxidation state in Roman numerals.

2 marks **Define a chelate and an ambidentate ligand with one example each.**

MODEL ANSWER

Chelate: a ring structure formed when a polydentate ligand binds to a metal through two or more donor atoms, e.g. ethylenediamine (en) forming a five-membered ring. It gives extra stability (chelate effect).

Ambidentate ligand: a monodentate ligand that can coordinate through either of two different donor atoms, e.g. NO_2^- (through N as nitrito-N or O as nitrito-O) and SCN^- (through S or N).

1 mark **Which is the stronger field ligand: H_2O or CN^-?**

MODEL ANSWER

CN^- (it is much higher in the spectrochemical series).

Ch 10 Haloalkanes and Haloarenes

HIGH

💡 Mechanisms are gold here. SN1 vs SN2, optical inversion, and the low reactivity of haloarenes are repeat questions.

IMPORTANT TOPICS (EXPLAINED)

Nomenclature & classification

Classify as 1°/2°/3° and as alkyl/allyl/benzyl/vinyl/aryl halides. Haloalkanes have $C(sp^3)$–X; haloarenes have $C(sp^2)$–X attached to an aromatic ring.

Preparation

From alcohols (with HX, PCl_3, PCl_5, $SOCl_2$ — $SOCl_2$ is best, by-products are gases), from alkenes (HX addition, Markovnikov; or halogenation), and Finkelstein/Swarts reactions (halide exchange).

Nucleophilic substitution — SN1 vs SN2

SN2: one step, bimolecular, rate $\propto$ [substrate][Nu]; back-side attack $\Longrightarrow$ **inversion of configuration (Walden inversion)**; favoured by primary halides + strong nucleophile + polar aprotic solvent.

SN1: two steps via carbocation, unimolecular, rate $\propto$ [substrate]; gives **racemisation**; favoured by tertiary halides + polar protic solvent. Reactivity SN1: 3° > 2° > 1°.

Elimination (β-elimination)

Dehydrohalogenation with alcoholic KOH gives alkenes; **Saytzeff rule** — the more substituted (more stable) alkene predominates.

Reactivity of haloarenes

Aryl halides are much **less reactive** towards nucleophilic substitution because of (i) resonance/partial double-bond character of C–X, (ii) sp^2 carbon (more electronegative, shorter stronger bond), and (iii) repulsion between the nucleophile and the electron-rich ring.

Polyhalogen compounds

Uses and concerns: **chloroform** ($CHCl_3$, forms poisonous phosgene on air-oxidation), **CCl_4**, **DDT** (persistent insecticide), **freons** (CFCs — ozone depletion).

MUST-KNOW FORMULAS & REACTIONS

```
Alcohol → halide (best): R–OH + SOCl₂ → R–Cl + SO₂ + HCl
```

```
SN2: rate = k[R–X][Nu], inversion; SN1: rate = k[R–X], racemisation
```

```
Reactivity SN1: 3° > 2° > 1° ; SN2: 1° > 2° > 3°
```

```
Dehydrohalogenation: R–CH2–CHX–R' + alc.KOH → alkene (Saytzeff)
```

HOW QUESTIONS COME FROM THIS CHAPTER

▸ 3-mark: distinguish SN1 and SN2 mechanisms (with rate law and stereochemistry).

▸ 2-mark: why are haloarenes less reactive than haloalkanes towards nucleophilic substitution?

▸ 2-mark: what is Walden inversion / Saytzeff rule?

▸ 1-mark: best reagent to convert an alcohol to a chloroalkane ($SOCl_2$).

PRACTICE QUESTIONS WITH MODEL ANSWERS

3 marks **Compare SN1 and SN2 reaction mechanisms with respect to molecularity, kinetics, stereochemistry and substrate preference.**

MODEL ANSWER

SN1 SN2

- **Steps/molecularity:** SN1 = two steps, unimolecular (carbocation intermediate); SN2 = one step, bimolecular.
- **Rate law:** SN1 rate = k[R–X]; SN2 rate = k[R–X][Nu].
- **Stereochemistry:** SN1 gives racemisation; SN2 gives inversion (Walden inversion).
- **Substrate:** SN1 favoured by 3° halides (stable carbocation); SN2 favoured by 1° halides (less steric hindrance).

2 marks **Why is chlorobenzene less reactive than chloromethane towards nucleophilic substitution?**

MODEL ANSWER

In chlorobenzene the C–Cl bond has **partial double-bond character** due to resonance of the chlorine lone pair into the ring, making it shorter and stronger. The carbon is also **sp^2 hybridised** (more electronegative, holds the bond tightly), and the electron-rich ring repels the incoming nucleophile. All three factors make substitution difficult, unlike in chloromethane.

2 marks **What is the Saytzeff rule? Illustrate with the dehydrohalogenation of 2-bromobutane.**

MODEL ANSWER

Saytzeff rule: in an elimination reaction, the alkene that is more highly substituted (and hence more stable) is formed as the major product. From 2-bromobutane with alcoholic KOH, **but-2-ene** (more substituted) is the major product over but-1-ene.

1 mark **Which reagent best converts ethanol to chloroethane with the cleanest by-products?**

MODEL ANSWER

Thionyl chloride ($SOCl_2$) — the by-products SO_2 and HCl are gases that escape, giving pure product.

Ch 11 Alcohols, Phenols and Ethers

HIGH

💡 Heavy organic chapter. Acidity of phenol, Reimer–Tiemann, Kolbe, Williamson synthesis and distinguishing tests are near-certain.

IMPORTANT TOPICS (EXPLAINED)

Preparation of alcohols & phenols

Alcohols: hydration of alkenes (Markovnikov / hydroboration-anti-Markovnikov), reduction of carbonyls, Grignard reagent + carbonyl. **Phenols:** from cumene (cumene → phenol + acetone), from chlorobenzene (Dow process), from diazonium salts.

Acidity of phenol vs alcohol

Phenol is far more acidic than alcohols because the **phenoxide ion is resonance-stabilised** (negative charge delocalised into the ring). Electron-withdrawing groups ($–NO_2$, especially ortho/para) increase acidity; electron-donating groups decrease it.

Reactions of alcohols

Esterification (with acids), dehydration (to alkenes, conc. H_2SO_4), oxidation (1° → aldehyde → acid; 2° → ketone; 3° resists), reaction with Na (H_2 evolved), Lucas test (distinguishes 1°/2°/3°).

Reactions of phenols

Electrophilic substitution is easy (–OH is activating, o/p-directing): bromination, nitration. **Kolbe's reaction** (phenol → salicylic acid via CO_2). **Reimer–Tiemann reaction** (phenol + $CHCl_3$ + KOH → salicylaldehyde). Reaction with $FeCl_3$ gives a violet colour (test for phenol).

Ethers — Williamson synthesis

Williamson ether synthesis: alkoxide + primary alkyl halide → ether (SN2; use primary halide to avoid elimination). Ethers undergo C–O cleavage with HI/HBr.

MUST-KNOW FORMULAS & REACTIONS

Williamson: $R\text{–}O^-Na^+ + R'\text{–}X \rightarrow R\text{–}O\text{–}R' + NaX$ (use 1° R'X)

Kolbe: sodium phenoxide + CO_2 (high P) → sodium salicylate → salicylic acid

Reimer–Tiemann: phenol + $CHCl_3$ + aq. KOH → salicylaldehyde (–CHO at ortho)

Oxidation: 1° alcohol → aldehyde → carboxylic acid; 2° alcohol → ketone

HOW QUESTIONS COME FROM THIS CHAPTER

- ▶ 3-mark: explain why phenol is more acidic than ethanol; effect of substituents.
- ▶ 2-mark: Reimer–Tiemann / Kolbe reaction with equation.
- ▶ 2-mark: Williamson synthesis — why use a primary halide?
- ▶ 2-mark: distinguish 1°, 2°, 3° alcohols (Lucas test) / test for phenol ($FeCl_3$).
- ▶ 1-mark: product of oxidation of a secondary alcohol (ketone).

PRACTICE QUESTIONS WITH MODEL ANSWERS

3 marks **Explain why phenol is more acidic than ethanol. How do nitro groups affect the acidity of phenol?**

MODEL ANSWER

On losing a proton, phenol gives the **phenoxide ion** in which the negative charge is delocalised over the ring by resonance, making it stable; ethanol gives the ethoxide ion where the charge is localised on oxygen and is destabilised by the electron-donating ethyl group. Greater anion stability $\Longrightarrow$ stronger acid, so phenol (pKa ≈ 10) is far more acidic than ethanol (pKa ≈ 16).

Nitro groups are electron-withdrawing; at the ortho/para positions they further delocalise the negative charge of the phenoxide ion, stabilising it. So nitrophenols are more acidic, and 2,4,6-trinitrophenol (picric acid) is a strong acid.

2 marks **Write the Reimer–Tiemann reaction with equation.**

MODEL ANSWER

When phenol is treated with chloroform ($CHCl_3$) and aqueous KOH at ~340 K, a –CHO group is introduced at the **ortho** position, giving **salicylaldehyde** (2-hydroxybenzaldehyde) after acid hydrolysis.

$C_6H_5OH + CHCl_3 + 3KOH \rightarrow o\text{-}HO–C_6H_4–CHO + 3KCl + 2H_2O$.

2 marks **Why is a primary alkyl halide preferred in Williamson's ether synthesis?**

MODEL ANSWER

Williamson synthesis proceeds by an **SN2** attack of the alkoxide on the alkyl halide. Alkoxides are strong bases as well as nucleophiles. With secondary or tertiary halides, the base promotes **elimination** (forming an alkene) instead of substitution. A primary halide has little steric hindrance, so substitution dominates and the ether forms in good yield.

1 mark **What colour does phenol give with neutral $FeCl_3$?**

MODEL ANSWER

A **violet/purple** colour (test for phenol).

Ch 12 Aldehydes, Ketones and Carboxylic Acids

HIGH

💡 Usually the single highest-weightage chapter. Named reactions and distinguishing tests (Tollens, Fehling, Iodoform) are guaranteed marks.

IMPORTANT TOPICS (EXPLAINED)

Nucleophilic addition

The carbonyl carbon (δ+) is attacked by nucleophiles. **Reactivity:** aldehydes > ketones (less steric hindrance, less +I from one R group). Addition of HCN (cyanohydrin), $NaHSO_3$, alcohols (acetal), ammonia derivatives (imines, oximes, hydrazones).

Aldol condensation

Carbonyl compounds with α-hydrogen, in dilute base, give a β-hydroxy carbonyl (aldol) which can dehydrate to an α,β-unsaturated carbonyl. Cross-aldol uses two different partners.

Cannizzaro reaction

Aldehydes with **no α-hydrogen** (e.g. HCHO, benzaldehyde) undergo disproportionation in concentrated base: one molecule is oxidised to the acid (salt) and another reduced to the alcohol.

Reduction reactions

Clemmensen (Zn-Hg/HCl) and **Wolff–Kishner** (NH_2NH_2/KOH) both reduce C=O to CH_2. $LiAlH_4$/$NaBH_4$ reduce carbonyls to alcohols.

Distinguishing tests

- **Tollens' reagent** (ammoniacal $AgNO_3$): aldehyde gives a silver mirror; ketone does not.
- **Fehling's solution**: aliphatic aldehyde gives a red-brown Cu_2O precipitate; ketone does not.
- **Iodoform test** (I_2/NaOH): positive for ethanal and methyl ketones (CH_3CO– group) and ethanol — yellow CHI_3 precipitate.

Carboxylic acids

Acidity is due to **resonance-stabilised carboxylate ion**; electron-withdrawing groups (Cl) increase acidity (e.g. trichloroacetic > acetic). Reactions: salt formation, esterification, conversion to acid chlorides/anhydrides/amides. **HVZ reaction** (α-halogenation with X_2/red P).

MUST-KNOW FORMULAS & REACTIONS

Tollens': $RCHO + 2[Ag(NH_3)_2]^+ + 3OH^- \rightarrow RCOO^- + 2Ag\downarrow$ (mirror) + ...

Iodoform: $CH_3CO{-}R$ (or $CH_3CH(OH)R$) $+ 3I_2 + 4NaOH \rightarrow CHI_3\downarrow + RCOONa$ + ...

Cannizzaro: $2HCHO + NaOH \rightarrow CH_3OH + HCOONa$ (no α-H)

Aldol: $2CH_3CHO$ --(dil.NaOH)--> $CH_3CH(OH)CH_2CHO \rightarrow CH_3CH{=}CHCHO + H_2O$

Acidity: $Cl_3C{-}COOH > Cl_2CH{-}COOH > ClCH_2{-}COOH > CH_3COOH$

HOW QUESTIONS COME FROM THIS CHAPTER

- ▶ 3-mark: distinguish given pairs by chemical tests (e.g. ethanal vs propanone, acetophenone vs benzophenone).
- ▶ 3-mark: write/explain Aldol, Cannizzaro, Clemmensen, Wolff–Kishner with equations.
- ▶ 2-mark: why are aldehydes more reactive than ketones towards nucleophilic addition?
- ▶ 2-mark: which compounds give a positive iodoform test?
- ▶ 1-mark: reagent for the silver-mirror test (Tollens').

PRACTICE QUESTIONS WITH MODEL ANSWERS

3 marks **How would you distinguish between (a) ethanal and propanone, and (b) propanal and propanone using chemical tests?**

MODEL ANSWER

(a) Ethanal (CH_3CHO) vs propanone (CH_3COCH_3): both give a positive iodoform test, so use **Tollens' reagent** — ethanal (an aldehyde) gives a silver mirror, propanone (a ketone) does not.

(b) Propanal vs propanone: use **Fehling's solution** — propanal (aldehyde) gives a red-brown precipitate of Cu_2O; propanone gives no reaction. Alternatively Tollens' (propanal → silver mirror).

3 marks **What is the Cannizzaro reaction? Write it for benzaldehyde and explain why benzaldehyde undergoes it but acetaldehyde does not.**

MODEL ANSWER

The **Cannizzaro reaction** is the base-induced disproportionation of an aldehyde that has **no α-hydrogen**: one molecule is oxidised to the carboxylate and another reduced to the alcohol.

$2C_6H_5CHO$ + conc. $NaOH \rightarrow C_6H_5COONa + C_6H_5CH_2OH$ (sodium benzoate + benzyl alcohol).

Benzaldehyde has no α-hydrogen, so it cannot undergo aldol condensation and instead disproportionates. Acetaldehyde has α-hydrogens, so it prefers **aldol condensation** and does not give the Cannizzaro reaction.

2 marks **Why are aldehydes more reactive than ketones towards nucleophilic addition?**

MODEL ANSWER

(1) **Electronic:** a ketone has two electron-donating alkyl/aryl groups that reduce the positive charge on the carbonyl carbon, whereas an aldehyde has only one, so the aldehyde carbon is more electrophilic. (2) **Steric:** the two bulky groups in a ketone hinder the approach of the nucleophile more than the one group in an aldehyde. Both effects make aldehydes react faster.

1 mark **Which reagent gives a silver mirror with aldehydes?**

MODEL ANSWER

Tollens' reagent (ammoniacal silver nitrate).

Ch 13 **Amines** HIGH

💡 Predictable scorer. Basicity order, preparation (Hoffmann, Gabriel), and diazonium-salt reactions appear most years.

IMPORTANT TOPICS (EXPLAINED)

Classification & preparation

1°, 2°, 3° amines and quaternary salts. Preparation: reduction of nitro compounds/nitriles/amides, ammonolysis of alkyl halides, **Gabriel phthalimide synthesis** (pure 1° amines only), **Hoffmann bromamide degradation** (amide → 1° amine with one fewer carbon).

Basicity of amines

Amines are basic (lone pair on N). In the gas phase basicity follows electron density, but in **aqueous solution** the order is governed by inductive effect + steric + solvation: typically 2° > 1° > 3° for many simple amines. **Aromatic amines (aniline) are weaker bases** than aliphatic amines because the N lone pair is delocalised into the ring.

Reactions of amines

Salt formation, acylation, reaction with nitrous acid (1° aliphatic → alcohol + N_2; 1° aromatic → diazonium salt), **carbylamine test** (1° amine + $CHCl_3$ + KOH → foul-smelling isocyanide — test for 1° amines), **Hinsberg's test** (distinguishes 1°/2°/3°).

Diazonium salts

Aryl diazonium salts ($Ar–N_2^+$) are versatile intermediates. **Sandmeyer/Gattermann** (replace N_2^+ by Cl, Br, CN), replacement by OH, H, I, F; **coupling reactions** with phenols/amines give azo dyes. They are key to making substituted aromatics that are hard to get directly.

MUST-KNOW FORMULAS & REACTIONS

```
Hoffmann bromamide: R–CONH₂ + Br₂ + 4NaOH → R–NH₂ + Na₂CO₃ + 2NaBr + 2H₂O (one C lost)
```

```
Gabriel: phthalimide → (KOH, R–X) → N-alkyl phthalimide → (hydrolysis) → R–NH₂ (pure 1°)
```

```
Diazotisation: Ar–NH₂ + NaNO₂ + 2HCl → Ar–N₂⁺Cl⁻ + NaCl + 2H₂O (0–5 °C)
```

```
Carbylamine (1° amine test): R–NH₂ + CHCl₃ + 3KOH → R–NC + 3KCl + 3H₂O
```

HOW QUESTIONS COME FROM THIS CHAPTER

- ▶ 3-mark: arrange amines in order of basic strength and justify (aniline vs methylamine etc.).
- ▶ 3-mark: synthetic uses of diazonium salts (Sandmeyer, coupling) with equations.
- ▶ 2-mark: Gabriel synthesis / Hoffmann bromamide degradation.
- ▶ 2-mark: distinguish 1°, 2°, 3° amines (Hinsberg / carbylamine).
- ▶ 1-mark: why can't aromatic amines be prepared by the Gabriel synthesis? (aryl halides don't undergo SN2).

PRACTICE QUESTIONS WITH MODEL ANSWERS

3 marks **Arrange the following in increasing order of basic strength in aqueous solution and justify: aniline, methylamine, ammonia.**

MODEL ANSWER

Order: aniline < ammonia < methylamine.

Aniline is the weakest: the nitrogen lone pair is delocalised into the benzene ring by resonance, so it is less available for donation. **Methylamine** is the strongest: the electron-donating (+I) methyl group increases electron density on nitrogen, making the lone pair more available and the conjugate acid more stable. Ammonia lies in between.

3 marks **Give the synthetic importance of diazonium salts with two example reactions.**

MODEL ANSWER

Aryl diazonium salts let us replace the $–N_2^+$ group with many groups that are hard to introduce directly, giving a wide range of substituted aromatics.

Sandmeyer reaction: $Ar–N_2^+Cl^-$ + CuCl → Ar–Cl + N_2 (also gives Ar–Br, Ar–CN with the right Cu salt).

Coupling reaction: $Ar–N_2^+$ + phenol → p-hydroxyazobenzene (an azo dye). These reactions are central to the manufacture of dyes.

2 marks **Describe the carbylamine test and state what it identifies.**

MODEL ANSWER

When a **primary amine** is heated with chloroform and alcoholic KOH, it forms an **isocyanide (carbylamine)** with an extremely offensive smell: $R–NH_2 + CHCl_3 + 3KOH \rightarrow R–NC + 3KCl + 3H_2O$. Secondary and tertiary amines do not give this smell, so the test specifically identifies primary amines.

1 mark **Which reaction converts an amide to a primary amine with one fewer carbon atom?**

MODEL ANSWER

Hoffmann bromamide degradation (with Br_2 + NaOH).

Ch 14 Biomolecules

MEDIUM

💡 Short and very scoring — pure memory. Glucose reactions, peptide bond, denaturation, DNA vs RNA are common 2–3 mark items.

IMPORTANT TOPICS (EXPLAINED)

Carbohydrates

Classified as **monosaccharides** (glucose, fructose), **disaccharides** (sucrose — non-reducing; maltose, lactose — reducing), **polysaccharides** (starch, cellulose, glycogen). Glucose is an aldohexose; it does not react with Schiff's reagent or $NaHSO_3$ (suggesting a cyclic hemiacetal form). Anomers (α/β), glycosidic linkage.

Proteins

Made of **α-amino acids** joined by **peptide (amide) bonds**. Amino acids exist as **zwitterions** at the isoelectric point. Structure levels: primary (sequence), secondary (α-helix, β-sheet via H-bonds), tertiary, quaternary. **Denaturation** destroys secondary/tertiary structure (heat, acid) — function lost, e.g. boiling an egg.

Enzymes & vitamins

Enzymes are biological catalysts (mostly proteins), highly specific. **Vitamins**: fat-soluble (A, D, E, K) and water-soluble (B group, C). Deficiency diseases (scurvy — C, rickets — D, night blindness — A).

Nucleic acids

DNA and **RNA** are polymers of nucleotides (base + sugar + phosphate). DNA: deoxyribose, bases A-T-G-C, double helix; carries genetic information. RNA: ribose, bases A-U-G-C, usually single-stranded; involved in protein synthesis.

MUST-KNOW FORMULAS & REACTIONS

```
Peptide bond: —CO—NH— (formed between —COOH of one amino acid and —NH₂ of the next, losing H₂O)
```

```
DNA bases pair: Adenine—Thymine, Guanine—Cytosine; RNA has Uracil instead of Thymine
```

```
Glucose: aldohexose, CHO—(CHOH)₄—CH₂OH; cyclic pyranose hemiacetal (α/β anomers)
```

HOW QUESTIONS COME FROM THIS CHAPTER

▸ 2-mark: differences between DNA and RNA.

▸ 2-mark: what is a peptide bond / zwitterion / denaturation of proteins?

▸ 2-mark: distinguish reducing and non-reducing sugars / why is sucrose non-reducing?

▸ 1-mark: which vitamin deficiency causes scurvy? (vitamin C).

PRACTICE QUESTIONS WITH MODEL ANSWERS

2 marks **Give two differences between DNA and RNA.**

MODEL ANSWER

- DNA contains the sugar **deoxyribose** and the base thymine; RNA contains **ribose** and uracil (instead of thymine).
- DNA is usually **double-stranded** (double helix) and stores genetic information; RNA is usually **single-stranded** and takes part in protein synthesis.

2 marks **What is denaturation of a protein? Give one example.**

MODEL ANSWER

Denaturation is the loss of the native secondary and tertiary structure of a protein (by heat, change in pH, or chemicals) while the primary structure (sequence) stays intact. The protein loses its biological activity. **Example:** coagulation of egg white (albumin) on boiling, or curdling of milk.

2 marks **Why is sucrose called a non-reducing sugar?**

MODEL ANSWER

In sucrose the reducing groups of glucose (its aldehyde/anomeric –OH) and fructose (its keto/anomeric –OH) are both involved in the **glycosidic linkage**. Because no free reducing (anomeric) group is available, sucrose cannot reduce Tollens' or Fehling's reagent, so it is non-reducing.

1 mark **Deficiency of which vitamin causes scurvy?**

MODEL ANSWER

Vitamin C (ascorbic acid).

Ch 15 Polymers

CORE

💡 Easy, descriptive marks. Classification, addition vs condensation, and naming the monomers of common polymers are frequent.

IMPORTANT TOPICS (EXPLAINED)

Classification of polymers

By **source:** natural (rubber, cellulose), semi-synthetic (cellulose acetate), synthetic (nylon). By **structure:** linear, branched, cross-linked. By **mode of polymerisation:** addition vs condensation. By **molecular forces:** elastomers, fibres, thermoplastics, thermosetting.

Addition vs condensation polymerisation

Addition (chain growth): unsaturated monomers add repeatedly with no loss of small molecules — polythene, PVC, polystyrene, Teflon (via free-radical/ionic mechanism). **Condensation** (step growth): bi-functional monomers combine with elimination of a small molecule (H_2O, HCl) — nylon-6,6, terylene, bakelite.

Important polymers & their monomers

- **Polythene** — ethene.
- **PVC** — vinyl chloride.
- **Teflon** — tetrafluoroethene.
- **Nylon-6,6** — hexamethylenediamine + adipic acid.
- **Terylene (Dacron)** — ethylene glycol + terephthalic acid.
- **Bakelite** — phenol + formaldehyde.
- **Natural rubber** — isoprene; vulcanisation (with S) hardens it.

Copolymers & biodegradable polymers

Copolymers form from two different monomers (e.g. Buna-S = butadiene + styrene). **Biodegradable polymers** decompose naturally — e.g. PHBV, Nylon-2-nylon-6 — reducing pollution.

MUST-KNOW FORMULAS & REACTIONS

Polythene: $n\ CH_2{=}CH_2 \rightarrow -(CH_2-CH_2)-_n$ (addition)

Nylon-6,6: hexamethylenediamine + adipic acid → polyamide + H_2O (condensation)

Terylene: ethylene glycol + terephthalic acid → polyester + H_2O

```
Bakelite: phenol + HCHO → cross-linked phenol-formaldehyde resin
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HOW QUESTIONS COME FROM THIS CHAPTER

- ▶ 2-mark: difference between addition and condensation polymers with examples.
- ▶ 2-mark: name the monomers of nylon-6,6 / terylene / bakelite / Buna-S.
- ▶ 1-mark: what is vulcanisation? / give one biodegradable polymer.
- ▶ 2-mark: classify polymers based on molecular forces (elastomer, fibre, thermoplastic, thermoset).

PRACTICE QUESTIONS WITH MODEL ANSWERS

2 marks **Distinguish between addition and condensation polymerisation with one example each.**

MODEL ANSWER

Addition polymerisation: unsaturated monomers join repeatedly without the loss of any small molecule; the polymer has the same empirical formula as the monomer. e.g. polythene from ethene.

Condensation polymerisation: two bi-functional monomers combine with the elimination of a small molecule such as water. e.g. nylon-6,6 from hexamethylenediamine and adipic acid.

2 marks **Name the monomers used to make (a) nylon-6,6 (b) terylene (c) bakelite.**

MODEL ANSWER

(a) Nylon-6,6 — **hexamethylenediamine and adipic acid**.

(b) Terylene — **ethylene glycol and terephthalic acid**.

(c) Bakelite — **phenol and formaldehyde**.

1 mark **What is vulcanisation of rubber?**

MODEL ANSWER

Heating natural rubber with **sulphur**, which forms cross-links between polymer chains, making the rubber harder, stronger and more elastic.

Ch 16 Chemistry in Everyday Life

CORE

💡 Pure memory, easy marks. Classes of drugs, soaps vs detergents and food additives are quick to revise and score.

IMPORTANT TOPICS (EXPLAINED)

Drugs & drug–target interaction

Drugs interact with biological targets (enzymes, receptors). **Enzyme inhibitors** block active sites; **receptor antagonists** block the natural messenger. Chemotherapy uses chemicals against disease.

Classes of medicines

- **Antacids** (reduce stomach acid — ranitidine, $Mg(OH)_2$).
- **Antihistamines** (allergy — relieve histamine effects).
- **Tranquilizers / antidepressants** (stress, anxiety).
- **Analgesics** (pain relief — aspirin; non-narcotic vs narcotic like morphine).
- **Antimicrobials / antibiotics** (penicillin) — broad/narrow spectrum, bactericidal/bacteriostatic.
- **Antiseptics** (applied to living tissue — Dettol) vs **disinfectants** (non-living surfaces — phenol).

Chemicals in food

Artificial sweeteners (saccharin, aspartame), **food preservatives** (sodium benzoate), antioxidants (BHT, BHA).

Cleansing agents — soaps vs detergents

Soaps are sodium/potassium salts of long-chain fatty acids; they don't work in hard water (form insoluble scum with Ca^{2+}/Mg^{2+}). **Synthetic detergents** work even in hard water; they may be anionic, cationic or non-ionic. Cleansing action: the hydrophobic tail dissolves grease, the hydrophilic head faces water, forming a **micelle** that is washed away.

MUST-KNOW FORMULAS & REACTIONS

```
Antiseptic vs disinfectant: antiseptic → living tissue (Dettol, dilute phenol); disinfectant → non-living surfaces (conc. phenol)
```

```
Soap: RCOO⁻Na⁺ (fails in hard water) ; Detergent: e.g. sodium lauryl sulphate (works in hard water)
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```
Cleansing action: micelle — hydrophobic tail into grease, hydrophilic head into water
```

HOW QUESTIONS COME FROM THIS CHAPTER

- ▸ 2-mark: difference between antiseptics and disinfectants with examples.
- ▸ 2-mark: difference between soaps and detergents / why soaps fail in hard water.
- ▸ 2-mark: explain the cleansing action of soap (micelle formation).
- ▸ 1-mark: give one example of an antacid / artificial sweetener / antibiotic.

PRACTICE QUESTIONS WITH MODEL ANSWERS

2 marks **Differentiate between antiseptics and disinfectants with one example each.**

MODEL ANSWER

Antiseptics are applied to living tissues (skin, wounds) to kill or prevent the growth of microorganisms, e.g. Dettol, dilute phenol, tincture of iodine.

Disinfectants are applied to non-living surfaces (floors, drains, instruments), e.g. concentrated phenol, chlorine, SO_2. The same substance can act as either, depending on concentration (e.g. phenol).

2 marks **Why do soaps not work well in hard water, and how do detergents solve this?**

MODEL ANSWER

Hard water contains Ca^{2+} and Mg^{2+} ions, which react with soap (sodium salts of fatty acids) to form insoluble **scum** (calcium/magnesium salts), wasting the soap and reducing lathering. **Synthetic detergents** are salts of sulphonic acids/sulphates whose calcium and magnesium salts are **soluble**, so they lather and clean even in hard water.

2 marks **Explain the cleansing action of soap.**

MODEL ANSWER

A soap molecule has a long hydrophobic (water-repelling) hydrocarbon tail and a hydrophilic (water-attracting) $–COO^-$ head. In water, the tails dissolve into the grease/oil on the dirty surface while the heads point outward into the water, forming a spherical **micelle** that traps the grease at its centre. These micelles are washed away with water, removing the dirt.

1 mark **Give one example of an antacid.**

MODEL ANSWER

Ranitidine (or magnesium hydroxide / sodium bicarbonate).

🔮 Predicted exam paper (high-probability)

DO THIS

💡 A full mock in the real Kerala 60-mark format, built from recurring past-paper patterns. The **% on each question is my honest estimate** of how likely that type appears — **not a leak**, just a guess from past papers. Questions are sorted **most-likely first**, so start at the top of each section. A 👉 start here tag marks the single best one to do first. Some questions show a ✏️ **diagram to draw** (drawing it in the exam earns marks). Answers are written the way you'd write them by hand. The SAY paper usually mirrors the regular March 2026 paper's pattern, so a 🔁 repeats yearly tag marks the types that come back almost every year. If you can do this paper, you pass.

SECTION A — 1 MARK EACH (Q1–5, ANSWER ANY 4)

Q1 · 1m ≈70% 👉 start here **Define molarity of a solution.**

MODEL ANSWER (EXAM STYLE)

Molarity (M) = number of moles of solute dissolved per litre of solution. It is temperature-dependent.

Q2 · 1m ≈70% **Name the reagent used in the silver-mirror test for aldehydes.**

MODEL ANSWER (EXAM STYLE)

Tollens' reagent (ammoniacal silver nitrate).

Q3 · 1m ≈65% **The half-life of a first-order reaction is independent of what?**

MODEL ANSWER (EXAM STYLE)

The initial concentration of the reactant.

Q4 · 1m ≈60% **What is the value of one Faraday of charge?**

MODEL ANSWER (EXAM STYLE)

96500 C — the charge carried by one mole of electrons.

Q5 · 1m ≈60% **Which is the stronger field ligand: H_2O or CN^-?**

MODEL ANSWER (EXAM STYLE)

CN^- (much higher in the spectrochemical series).

SECTION B — 2 MARKS EACH (Q6–15, ANSWER ANY 8)

Q6 · 2m ≈80% 🔁 repeats yearly 👉 start here **Distinguish between order and molecularity of a reaction.**

MODEL ANSWER (EXAM STYLE)

Order = sum of powers of concentration terms in the experimental rate law; can be zero/fractional/integral, defined for the overall reaction. Molecularity = number of species in an elementary step; always a whole number, defined only for elementary reactions.

Q7 · 2m ≈80% 🔁 repeats yearly **Why is phenol more acidic than ethanol?**

MODEL ANSWER (EXAM STYLE)

The phenoxide ion formed after losing H^+ is resonance-stabilised (charge delocalised into the ring), whereas the ethoxide ion has the charge localised on oxygen and destabilised by the +I ethyl group. Greater anion stability $\Longrightarrow$ stronger acid.

Q8 · 2m ≈80% 🔁 repeats yearly **Arrange in increasing order of basic strength in water: aniline, ammonia, methylamine.**

MODEL ANSWER (EXAM STYLE)

aniline < ammonia < methylamine. Aniline's lone pair is delocalised into the ring (least available); methylamine's +I methyl group raises electron density on N (most basic).

Q9 · 2m ≈80% 🔁 repeats yearly **What is lanthanoid contraction and one consequence?**

MODEL ANSWER (EXAM STYLE)

The steady decrease in atomic/ionic radii across the 4f series due to poor shielding by 4f electrons. Consequence: the second and third transition series of a group have almost identical sizes (e.g. Zr and Hf), so they are hard to separate.

Q10 · 2m ≈75% **Define a chelate and an ambidentate ligand with one example each.**

MODEL ANSWER (EXAM STYLE)

Chelate: ring formed when a polydentate ligand binds a metal through two or more donor atoms (e.g. ethylenediamine). Ambidentate ligand: a ligand that can bind through either of two donor atoms (e.g. NO_2^- via N or O; SCN^- via S or N).

Q11 · 2m ≈75% **Why are aldehydes more reactive than ketones towards nucleophilic addition?**

MODEL ANSWER (EXAM STYLE)

A ketone has two electron-donating alkyl groups that lower the positive charge on the carbonyl carbon and sterically hinder attack; an aldehyde has only one, so its carbonyl carbon is more electrophilic and less crowded — hence more reactive.

Q12 · 2m ≈75% **Distinguish between addition and condensation polymerisation with one example each.**

MODEL ANSWER (EXAM STYLE)

Addition: unsaturated monomers add repeatedly with no loss of small molecules (e.g. polythene from ethene). Condensation: bi-functional monomers combine with elimination of a small molecule like water (e.g. nylon-6,6).

Q13 · 2m ≈70% **State Henry's law and give one application.**

MODEL ANSWER (EXAM STYLE)

The solubility of a gas in a liquid is directly proportional to its partial pressure: $p = K_H \cdot x$. Application: aerated/soft drinks are bottled under high CO_2 pressure to increase dissolved gas.

Q14 · 2m ≈70% **State Kohlrausch's law and give one use.**

MODEL ANSWER (EXAM STYLE)

At infinite dilution, the limiting molar conductivity equals the sum of the individual ionic contributions. Use: finding $\Lambda°_m$ of a weak electrolyte (e.g. acetic acid) and its degree of dissociation.

Q15 · 2m ≈70% **Give two differences between DNA and RNA.**

MODEL ANSWER (EXAM STYLE)

DNA has deoxyribose sugar and the base thymine; RNA has ribose and uracil. DNA is usually double-stranded and stores genetic information; RNA is usually single-stranded and helps in protein synthesis.

SECTION C — 3 MARKS EACH (Q16–26, ANSWER ANY 8)

Q16 · 3m

≈85%

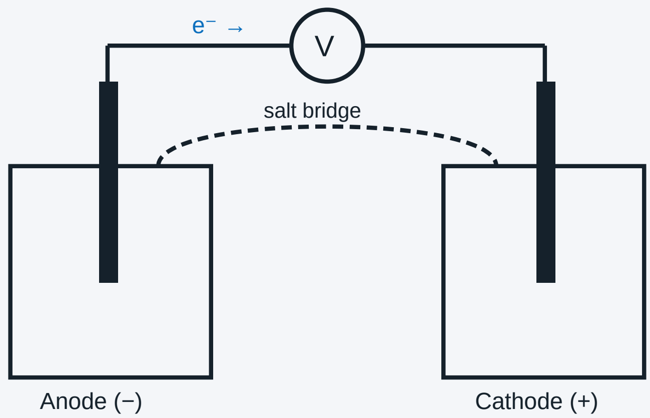
🔁 repeats yearly

👉 start here

Calculate the EMF of Zn|Zn^{2+}(0.1 M)||Cu^{2+}(1.0 M)|Cu at 298 K. E°(Zn^{2+}/Zn)=−0.76 V, E°(Cu^{2+}/Cu)=+0.34 V.

MODEL ANSWER (EXAM STYLE)

> ✏️ **Draw this:** a galvanic cell — two beakers, an electrode in each, a salt bridge on top and a voltmeter; mark anode (−) and cathode (+).



Galvanic cell. Electrons flow through the wire from the anode (−) to the cathode (+); ions move through the salt bridge.

$E°_{cell} = 0.34 - (-0.76) = 1.10$ V. n=2, Q = $[Zn^{2+}]/[Cu^{2+}]$ = 0.1. E = 1.10 − (0.0591/2)·log(0.1) = 1.10 + 0.0296 = 1.13 V.

Q17 · 3m

≈85%

🔁 repeats yearly

A first-order reaction has k = 1.15×10^{-3} s^{-1}. Time for 5 g reactant to fall to 3 g?

MODEL ANSWER (EXAM STYLE)

$t = (2.303/k)\cdot\log([R]_0/[R]) = (2.303/1.15\times10^{-3})\cdot\log(5/3) = 2002.6 \times 0.2219 \approx 444$ s.

Q18 · 3m

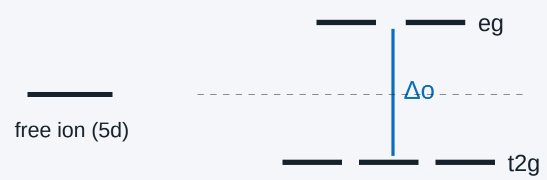
≈85%

🔁 repeats yearly

Using CFT, explain why $[Co(NH_3)_6]^{3+}$ is diamagnetic but $[CoF_6]^{3-}$ is paramagnetic.

MODEL ANSWER (EXAM STYLE)

✏️ **Draw this:** the d-orbital splitting: a lower t2g set (3 lines) and an upper eg set (2 lines) with the Δo gap between them.



Octahedral d-orbital splitting. Strong-field ligand (CN^-, NH_3) → big Δo → electrons pair in t2g → low-spin. Weak-field (F^-) → small Δo → high-spin.

Both are Co^{3+} (d^6), octahedral. NH_3 is strong-field → large Δ_o → electrons pair as $t_{2g}^6\ e\_g^0$, no unpaired electrons → diamagnetic (low-spin). F^- is weak-field → small Δ_o → $t_{2g}^4\ e\_g^2$, 4 unpaired electrons → paramagnetic (high-spin).

Q19 · 3m

≈85%

🔁 repeats yearly

How would you distinguish (a) ethanal & propanone and (b) propanal & propanone by chemical tests?

MODEL ANSWER (EXAM STYLE)

(a) Both give iodoform; use Tollens' — ethanal gives a silver mirror, propanone does not. (b) Use Fehling's/Tollens' — propanal (aldehyde) gives red-brown Cu_2O / silver mirror; propanone gives no reaction.

Q20 · 3m

≈80%

🔁 repeats yearly

Write IUPAC names: (a) $K_4[Fe(CN)_6]$ (b) $[Cr(NH_3)_6]Cl_3$ (c) $[Co(en)_3]Cl_3$.

MODEL ANSWER (EXAM STYLE)

(a) Potassium hexacyanidoferrate(II). (b) Hexaamminechromium(III) chloride. (c) Tris(ethane-1,2-diamine)cobalt(III) chloride.

Q21 · 3m

≈80%

🔁 repeats yearly

Compare SN1 and SN2 mechanisms (molecularity, rate law, stereochemistry, substrate).

MODEL ANSWER (EXAM STYLE)

SN1: two steps via carbocation, unimolecular, rate = k[RX], racemisation, favoured by 3° halides. SN2: one step, bimolecular, rate = k[RX][Nu], inversion (Walden), favoured by 1° halides.

Q22 · 3m

≈80%

🔁 repeats yearly

Give the synthetic importance of diazonium salts with two example reactions.

MODEL ANSWER (EXAM STYLE)

They let $–N_2^+$ be replaced by groups hard to add directly. Sandmeyer: $Ar–N_2^+$ + CuCl → Ar–Cl + N_2. Coupling: $Ar–N_2^+$ + phenol → p-hydroxyazobenzene (azo dye).

Q23 · 3m

≈75%

1.0 g of a non-electrolyte in 50 g benzene lowers freezing point by 0.40 K (K_f = 5.12). Find its molar mass.

MODEL ANSWER (EXAM STYLE)

M = (K_f·w_B·1000)/(ΔT_f·w_A) = (5.12×1.0×1000)/(0.40×50) = 256 g/mol.

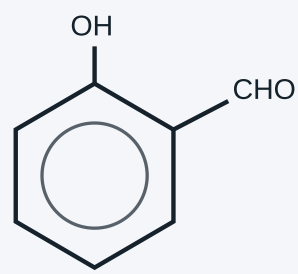
Q24 · 3m

≈75%

Explain why phenol is more acidic than ethanol and how a nitro group affects phenol's acidity.

MODEL ANSWER (EXAM STYLE)

✏️ **Draw this:** the benzene ring with –OH; for the Reimer–Tiemann part put –OH and –CHO on next-door (ortho) carbons.



Salicylaldehyde — the –OH and –CHO sit on neighbouring (ortho) carbons. This is the Reimer–Tiemann product of phenol.

Phenoxide is resonance-stabilised (charge into ring) so phenol is more acidic than ethanol. Electron-withdrawing $–NO_2$ at ortho/para further delocalises the negative charge, increasing acidity (picric acid is strongly acidic).

Q25 · 3m

≈75%

Calculate the spin-only magnetic moment of Fe^{2+} and explain why transition metals show variable oxidation states.

MODEL ANSWER (EXAM STYLE)

$Fe^{2+} = 3d^6$, n = 4 unpaired; $\mu = \sqrt{(4\times6)} = 4.90$ BM. Variable oxidation states arise because (n−1)d and ns energies are close, so different numbers of electrons can take part in bonding.

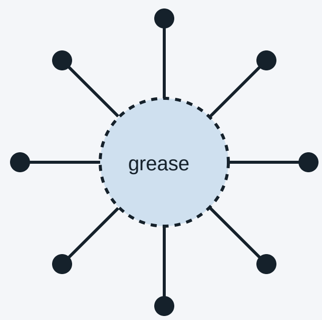
Q26 · 3m

≈75%

Explain the cleansing action of soap, and distinguish antiseptics from disinfectants.

MODEL ANSWER (EXAM STYLE)

✏️ **Draw this:** a micelle: a grease drop in the centre with soap molecules around it — tails pointing in, round ionic heads pointing out into water.



Soap micelle. The oil-loving tails point into the grease; the water-loving heads (●) point outward, so the whole ball lifts off and washes away.

Soap's hydrophobic tail dissolves in grease while the hydrophilic head faces water, forming a micelle that traps dirt and is washed away. Antiseptics are applied to living tissue (Dettol); disinfectants to non-living surfaces (conc. phenol).

SECTION D — 4 MARKS EACH (Q27–31, ANSWER ANY 4)

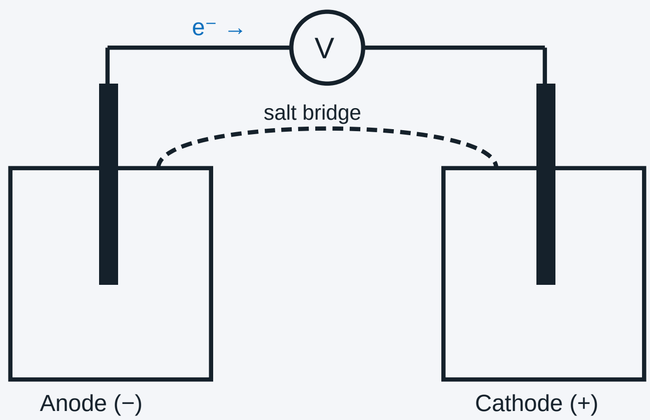
Q27 · 4m

≈80% 🔁 repeats yearly 👉 start here

For the cell $Mg|Mg^{2+}(0.1\ M)||Cu^{2+}(0.01\ M)|Cu$, write the cell reaction, find E°_cell, E_cell and ΔG°. [E°(Mg^{2+}/Mg)=−2.37 V, E°(Cu^{2+}/Cu)=+0.34 V]

MODEL ANSWER (EXAM STYLE)

✏️ **Draw this:** the galvanic cell (two beakers, electrodes, salt bridge, voltmeter).



Galvanic cell. Electrons flow through the wire from the anode (−) to the cathode (+); ions move through the salt bridge.

Reaction: $Mg + Cu^{2+} \rightarrow Mg^{2+} + Cu$. E°_cell = 0.34 −(−2.37) = 2.71 V. n=2, Q = $[Mg^{2+}]/[Cu^{2+}]$ = 0.1/0.01 = 10. E = 2.71 − (0.0591/2)·log10 = 2.71 − 0.0296 = 2.68 V. ΔG° = −nFE° = −2×96500×2.71 = −523 kJ (≈ -5.23×10^{5} J).

Q28 · 4m

≈80% 🔁 repeats yearly

Using crystal field theory, explain the colour and magnetic behaviour of coordination compounds. Define the spectrochemical series with one strong and one weak ligand.

MODEL ANSWER (EXAM STYLE)

✏️ **Draw this:** the octahedral splitting (t2g, eg, Δo) and show one electron jumping t2g→eg to explain colour.

Octahedral d-orbital splitting. Strong-field ligand (CN^-, NH_3) → big Δo → electrons pair in t2g → low-spin. Weak-field (F^-) → small Δo → high-spin.

Ligands split d-orbitals (octahedral: t_2g below, e_g above, gap Δ_0). An electron absorbs visible light to jump t_2g→e_g; the complementary colour is seen (colour). Number of unpaired electrons after filling decides magnetism (paramagnetic if unpaired). The spectrochemical series ranks ligands by field strength — weak: I^-, F^-, H_2O; strong: NH_3, en, CN^-, CO. Strong-field → low-spin; weak-field → high-spin.

Q29 · 4m

≈80% 🔁 repeats yearly

Complete and name the reactions: (a) sodium phenoxide + CO_2 (b) phenol + $CHCl_3$ + KOH (c) alkoxide + primary alkyl halide (d) salicylic acid use.

MODEL ANSWER (EXAM STYLE)

(a) Kolbe's reaction → sodium salicylate → salicylic acid. (b) Reimer–Tiemann → salicylaldehyde (–CHO at ortho). (c) Williamson ether synthesis → ether (SN2; primary halide avoids elimination). (d) Salicylic acid is the basis of aspirin (acetylsalicylic acid).

Q30 · 4m

≈80%

🔁 repeats yearly

Describe the preparation of a primary amine by (a) Gabriel synthesis and (b) Hoffmann bromamide degradation, and describe the carbylamine test for primary amines.

MODEL ANSWER (EXAM STYLE)

(a) Gabriel: phthalimide → (KOH, R–X) → N-alkylphthalimide → hydrolysis → pure 1° amine (no aromatic amines, as aryl halides don't do SN2). (b) Hoffmann: $R–CONH_2 + Br_2 + 4NaOH \rightarrow R–NH_2$ (one carbon fewer). Carbylamine test: 1° amine + $CHCl_3$ + alc.KOH → foul-smelling isocyanide (R–NC) — confirms a primary amine.

Q31 · 4m

≈70%

What is the Cannizzaro reaction? Write it for benzaldehyde and explain why benzaldehyde undergoes it but acetaldehyde does not.

MODEL ANSWER (EXAM STYLE)

Cannizzaro = base-induced disproportionation of an aldehyde with no α-hydrogen: one molecule is oxidised to the carboxylate, another reduced to the alcohol. $2C_6H_5CHO$ + conc. NaOH → $C_6H_5COONa + C_6H_5CH_2OH$. Benzaldehyde has no α-H so it disproportionates; acetaldehyde has α-H and prefers aldol condensation instead.

📋 Exam pattern — how the 60 marks are split

- **Total questions: 31.** You don't answer all of them — there is internal choice. Read instructions on the day.
- **Q1–5 · 1 mark each** (objective / fill-in / match / one-word) — attempt any 4 → 4 marks.
- **Q6–15 · 2 marks each** (very short) — attempt any 8 → 16 marks.
- **Q16–26 · 3 marks each** (short) — attempt any 8 → 24 marks.
- **Q27–31 · 4 marks each** (long) — attempt any 4 → 16 marks.
- **Theory total = 60.** Internal/practical = 40. Grand total = 100.
- **You must score at least 18/60 (30%) in theory to pass.** There is **no negative marking**, so attempt everything.

✍️ How to write answers and grab easy marks

- **Always draw the structure / equation.** In organic questions a correct balanced equation alone can fetch most of the marks even if your words are weak.
- **Balance every chemical equation** and write conditions over the arrow (catalyst, temperature, reagent). Examiners give a separate mark for conditions.
- **Numericals: write the formula first, substitute, then answer with units.** You get step marks even if the final number is wrong. Never skip the formula line.
- **Definitions: give the one-line definition + one example.** "Define + example" pattern wins full 2-mark questions.
- **Distinguishing tests: name the reagent + state what each compound does** (e.g. "aldehyde gives silver mirror with Tollens', ketone does not").
- **Use keywords.** Markers look for terms like "nucleophilic addition", "lanthanoid contraction", "colligative", "spectrochemical series". Sprinkle them in.
- **Attempt the choice question you know best.** With internal choice, pick the version where you can draw a clean diagram.
- **Time: ~2 min per mark.** Don't over-write a 1-mark answer. Bullet points are fine and faster to mark.

Sources used to build this (official syllabus + exam pattern):

Kerala Plus Two Syllabus & mark distribution — Careers360 · DHSE Kerala exam pattern & marking scheme — Careers360 · Chapter-wise weightage — HSSLive · Plus Two Chemistry chapter notes — A Plus Topper

Built as original revision material from the standard Kerala/NCERT Class 12 syllabus. Verify the live mark-scheme on dhsekerala.gov.in on exam day. Progress (ticked chapters, exam time) is saved in your browser only. Good luck — you've got this. ⚗️